

The Global Matrix Method and the Source Jump

The stratified reference solver, and $\delta(z - z_1) \mathbf{I}_6$ as its source primitive

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Abstract

The stratified reference medium of the multiple-scattering programme is solved by a global matrix: one block-tridiagonal system per horizontal wavenumber and frequency, assembled from layer eigenvectors and interface continuity, and reduced to an $O(N)$ sweep by a block-Riccati recursion. This note states that construction as implemented, and then addresses the part of it that has never been written down completely — the source. A source enters the global matrix as a *jump* in the six-component state across its own plane. We show that the up- and down-going radiation produced by the unit jump $\delta(z - z_1) \mathbf{I}_6$ is exactly the inverse eigenvector matrix, available in closed form without a linear solve, because the layer modes are symplectically orthogonal. One matrix therefore yields every column of the Green’s function at once. The existing treatment restricts the source to the free-surface layer or the lower half-space; the unit jump removes that restriction.

Companion series. *One of a companion series on elastic multiple scattering from cubic heterogeneities on a depth-stratified background, the T-matrix programme $T = T_0(I - G_0 T_0)^{-1}$. This note covers the reference solver: the stratified medium on which the scattering series is built. Its companions treat the inter-site propagator, the contrast-source volume integral equation, and the directional partial-wave sweep matvec.*

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1 The stratified system

Work at fixed angular frequency ω and fixed horizontal wavenumber $\mathbf{k}_{\parallel} = (k_x, k_y)$, with the time convention $e^{-i\omega t}$, depth z positive downward, and seismic units (km s^{-1} , g cm^{-3} , GPa , km). The medium is a stack of homogeneous layers; within each, the displacement and traction satisfy a first-order system in z whose solutions are plane waves.

The natural dependent variable is the six-component *state*

$$\mathbf{v}(z) = (u_z, u_x, u_y, \tau_{zz}, \tau_{xz}, \tau_{yz})^T, \quad (1)$$

because exactly these six quantities are continuous across a welded interface. Everything else — the in-plane strains, the remaining stress components — jumps with the material, and any formulation carrying them across an interface has to undo that jump again.

In each layer the system supports six modes: P , SV and SH , each up- or down-going. Collecting their state vectors as the columns of a matrix $\mathbf{D}_z(\mathbf{k}_{\parallel})$, the field in layer j , spanning $z_j \leq z \leq z_{j+1}$, is

$$\mathbf{v}(z) = \mathbf{D}_z^{(j)} \begin{pmatrix} e^{+i\omega\eta(z-z_j)} \mathbf{c}_{\downarrow} \\ e^{-i\omega\eta(z-z_{j+1})} \mathbf{c}_{\uparrow} \end{pmatrix}, \quad (2)$$

with η the vertical slowness of the mode in question, on the branch $\text{Im } \eta \geq 0$.

Each amplitude is referenced to the boundary its wave starts from. This is not cosmetic, and it is the single most important convention in the whole construction. The down-going amplitude $\mathbf{c}_{\downarrow}$ is referenced at the *top* z_j and the up-going amplitude $\mathbf{c}_{\uparrow}$ at the *bottom* z_{j+1} , so that both exponents in (2) have the form $+i\omega\eta \times (\text{positive distance travelled})$. Writing $\eta = \eta_r + i\eta_i$ with $\eta_i \geq 0$, each factor is then

$$e^{i\omega\eta_r d} e^{-\omega\eta_i d}, \quad d > 0,$$

which *decays*. Referencing the up-going wave to the top instead — the apparently symmetric choice $e^{-i\omega\eta(z-z_j)}$ — gives $e^{+\omega\eta_i d}$ and builds a growing exponential into the formulation. For propagating wavenumbers the difference is a bookkeeping convention; for evanescent ones, where η_i is large and $\omega\eta_i h$ may be tens or hundreds, it is the difference between a stable recursion and an overflow. The growing exponential is never formed anywhere in this construction, and (2) is where that begins.

The amplitudes $\mathbf{c}_{\downarrow}, \mathbf{c}_{\uparrow}$ are the unknowns. Every phase factor that appears below is consequently $e^{+i\omega\eta h}$ with $h > 0$ a layer thickness or a fraction of one: a single layer crossing, always in the decaying direction.

Decoupling. At fixed $\mathbf{k}_{\parallel}$ the six modes separate into a four-component P – SV system in the sagittal plane and a two-component SH system out of it, related to the Cartesian state by a rotation through the azimuth $\phi = \arctan(k_y/k_x)$. The implementation carries the two blocks separately and rotates on output. The P – SV state is $(u_x, u_z, \tau_{zz}/(-i\omega), \tau_{xz}/(-i\omega))$ and the SH state is $(u_t, \tau_{tz}/(-i\omega))$; the $(-i\omega)$ scaling keeps the layer matrices frequency-independent.

2 The global matrix

Continuity of $\mathbf{v}$ at every interface, together with the free surface above and the radiation condition below, closes the system. Writing the layer amplitudes as the unknown vector and continuity as the equations, the result is one linear system per $(\omega, \mathbf{k}_{\parallel})$,

$$\mathcal{A} \mathbf{c} = \mathbf{b}, \quad (3)$$

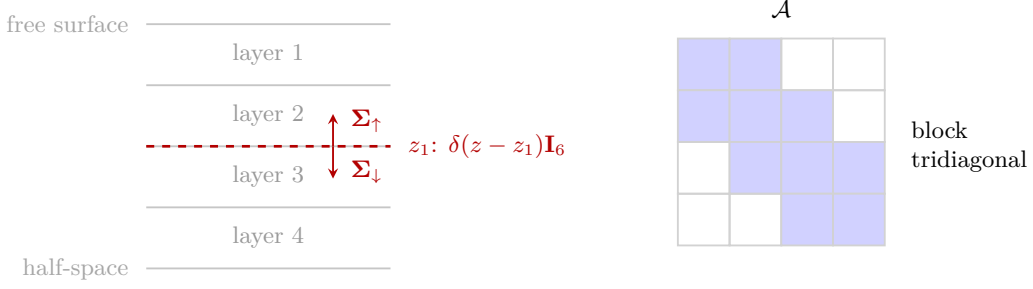


Figure 1: Left: the layer stack, with a source plane at z_1 radiating $\Sigma_\uparrow$ upward and $\Sigma_\downarrow$ downward. Right: the global matrix $\mathcal{A}$ of (3). Each interface couples only neighbouring layers, so $\mathcal{A}$ is block tridiagonal — which is what the Riccati recursion of §3 exploits.

whose matrix is *block tridiagonal*: interface k couples only the amplitudes of the layers immediately above and below it.

Solving (3) densely costs $O(N^3)$ in the number of layers, and it must be solved afresh for every frequency and every wavenumber. That cost is the reason for the next section.

3 The Riccati reduction

Block tridiagonality permits the standard elimination: sweep from the half-space upward, carrying at each interface a single 2×2 matrix $\mathbf{Y}_k$ that summarises everything below it. At interface k , with $\mathbf{E}_d$ and $\mathbf{E}_u$ the down- and up-going eigenvector blocks and $\mathbf{E}$ the layer phase,

$$\begin{bmatrix} \mathbf{E}_u^{(k)} & -\mathbf{E}_{\text{eff}}^{(k+1)} \end{bmatrix} \begin{pmatrix} \mathbf{Y}_k \\ * \end{pmatrix} = -\mathbf{E}_d^{(k)} \mathbf{E}^{(k)}, \quad \mathbf{E}_{\text{eff}}^{(k+1)} = \mathbf{E}_d^{(k+1)} + \mathbf{E}_u^{(k+1)} \mathbf{E}^{(k+1)} \mathbf{Y}_{k+1}, \quad (4)$$

so each step is a 4×4 solve and the whole sweep is $O(N)$. $\mathbf{Y}_k$ is the reflection matrix of the stack below interface k , and the recursion is the matrix Riccati equation of invariant embedding.

Its stability is inherited directly from (2) rather than imposed. Because $\mathbf{c}_\downarrow$ is referenced at the top of its layer and $\mathbf{c}_\uparrow$ at the bottom, the phase $\mathbf{E}^{(k)}$ multiplying $\mathbf{E}_d^{(k)}$ in (4) carries the down-going wave from the top to the bottom, and the phase multiplying $\mathbf{E}_u^{(k+1)}$ carries the up-going wave from the bottom to the top. Both are $e^{+i\omega\eta h}$ with $h > 0$; both decay. No step of the sweep ever forms a growing exponential, which is what lets it run to arbitrary evanescence without conditioning collapse.

A particular solution rides alongside the homogeneous sweep. Writing $\mathbf{g}_k$ for the accumulated source contribution below interface k , the same factorisation that gives $\mathbf{Y}_k$ gives $\mathbf{g}_k$ at no extra cost: the right-hand side gains a column, and the 4×4 solve is performed once against both. Sources therefore cost one extra column, not one extra sweep.

4 The source as a jump

Now the part that motivates this note.

A source does not enter (3) as a right-hand side in the obvious place. It enters as a *discontinuity* in the state across its own plane: on either side of z_1 the field is source-free and satisfies (2), and the source is the mismatch between the two solutions,

$$\mathbf{v}(z_1^+) - \mathbf{v}(z_1^-) = \mathbf{S}. \quad (5)$$

Everything about a source is in that six-vector. A body force, a moment tensor, an explosion and a stress glut differ only in the $\mathbf{S}$ they produce.

The unit jump. It is therefore natural to take as the primitive not any particular physical source but the *unit* jump

$$\mathbf{S} = \delta(z - z_1) \mathbf{I}_6, \quad (6)$$

whose six columns are the six independent state discontinuities. The response to (6) is the Green's function of the stratified medium: all six source components at once, one object rather than six solves. Any physical source is then a linear combination, formed *after* the expensive part is done.

4.1 Decomposing the jump: symplectic orthogonality

To use (6) the jump must be split into the amplitudes it excites. With $\mathbf{E} = [\mathbf{E}_d \mid \mathbf{E}_u]$ the full eigenvector matrix of the source layer, the amplitudes $\mathbf{c}$ solve $\mathbf{E}\mathbf{c} = \mathbf{S}$, so for the unit jump

$$\mathbf{c} = \mathbf{E}^{-1}. \quad (7)$$

That looks like a matrix inversion per frequency and per wavenumber. It is not, and the reason is structural.

Define the symplectic form on the state,

$$\langle \mathbf{b}_1, \mathbf{b}_2 \rangle = \mathbf{b}_1^\top \mathbf{J} \mathbf{b}_2, \quad \mathbf{J} = \begin{pmatrix} \mathbf{0} & \mathbf{I}_3 \\ -\mathbf{I}_3 & \mathbf{0} \end{pmatrix}, \quad (8)$$

pairing each displacement component with its conjugate traction. The elastic layer operator is Hamiltonian with respect to (8), and its eigenvectors are consequently *symplectically orthogonal*: the Gram matrix

$$\mathbf{G} = \mathbf{E}^\top \mathbf{J} \mathbf{E} \quad (9)$$

is block-antidiagonal with *diagonal* blocks. Each mode pairs with its own opposite-going partner and with nothing else — $P_\downarrow$ with $P_\uparrow$, $S_\downarrow$ with $S_\uparrow$, never P with S . From (9) the inverse follows immediately:

$$\boxed{\mathbf{E}^{-1} = \mathbf{G}^{-1} \mathbf{E}^\top \mathbf{J}} \quad (10)$$

and $\mathbf{G}^{-1}$ is a block swap together with one reciprocal per mode. The inversion has disappeared.

The same statement, in the energy-normalised basis. When the eigenvectors are scaled so that the pairings in (9) are unity, the Gram matrix becomes $\mathbf{J}$ itself and (10) reduces to

$$\mathbf{D}_z^{-1}(\mathbf{k}_\parallel) = -i \mathbf{J} \mathbf{D}_z^\top(-\mathbf{k}_\parallel) \mathbf{J}, \quad (11)$$

the symplectic inverse. Equation (10) is that identity written for eigenvectors in whatever normalisation they happen to carry: the structure survives any diagonal rescaling, so no special normalisation is required to use it.

4.2 What this replaces

The existing treatment computes the up- and down-going radiation from a dipolar body force $\mathbf{F} = \mathbf{F}_1 \delta(z - z_S) + \mathbf{F}_2 \delta'(z - z_S)$ as

$$\begin{pmatrix} \Sigma_\uparrow \\ \Sigma_\downarrow \end{pmatrix} = -i \langle \mathbf{D}_{z,S}(-\mathbf{k}_\parallel), \mathbf{F}_1 + \mathbf{A}_S \mathbf{F}_2 \rangle, \quad (12)$$

and then restricts attention to a source in the free-surface layer or the lower half-space, for which the assembled right-hand side $\mathbf{b}$ has a closed form.

Equations (6)–(10) are (12) with $\mathbf{F}_1 = \mathbf{I}_6$ and $\mathbf{F}_2 = \mathbf{0}$, and they carry two consequences. The source may sit at *any* depth, interior layers included, because nothing in the derivation referred to the boundaries. And the combination $\mathbf{F}_1 + \mathbf{A}_S \mathbf{F}_2$ need not be formed at all when what is wanted is the Green's function itself: the unit jump is the identity, and a particular physical source is applied afterwards as a linear combination of columns.

4.3 Placing the radiation

With $\mathbf{c} = \mathbf{E}^{-1}$ in hand, the down-going half is propagated to the bottom of the source layer and the up-going half to the top, each by its own phase factor, and the two enter the Riccati right-hand side at the interfaces bounding the layer:

$$\Sigma_{\uparrow}^{(k-1)} = \mathbf{E}_u e^{i\omega\eta fh} \mathbf{c}_{\uparrow}, \quad \Sigma_{\downarrow}^{(k)} = -\mathbf{E}_d e^{i\omega\eta(1-f)h} \mathbf{c}_{\downarrow}, \quad (13)$$

with h the layer thickness and f the fractional depth of the source within it. The sign on $\Sigma_{\downarrow}$ is the side of (5) it appears on. Both propagations run in the decaying direction, so (13) inherits the stability of the sweep.

5 The marine stack: free surface, ocean, layers, half-space

The stack that matters in practice is not welded throughout. It is bounded above by a *free surface*, below that a *fluid* ocean layer of finite thickness, then the elastic layers, and finally a radiating half-space. Two of those four differ structurally from §1, and both change the boundary conditions rather than merely the coefficients.

The fluid layer carries two modes, not six. A fluid supports no shear, so $\beta = 0$ and the *SV* and *SH* columns of $\mathbf{D}_z$ are absent: the state reduces to (u_z, τ_{zz}) and the layer propagates a single up-going and a single down-going *P* amplitude. At the seabed the welded condition of (1) is therefore the wrong one. Only u_z and τ_{zz} are continuous; the tangential traction τ_{xz} must *vanish* on the fluid side, and the tangential displacement u_x is free to slip. The interface is an acoustic-elastic one, with a reflection matrix whose $\mathbf{R}_d$ has a single non-zero entry and whose transmission operators are rank-deficient — the fluid cannot receive an *SV* wave, so one column of $\mathbf{T}_d$ and one row of $\mathbf{T}_u$ are identically zero.

The free surface is a pressure release. At the top of the water column the traction vanishes identically, so the up-going amplitude is reflected with $\mathbf{R}_{fs} = -1$. The reverberation it closes is a geometric series in the two-way water delay: writing $\mathbf{R}$ for the response referenced at the top of the water, already carrying one down-and-up passage of the ocean,

$$\mathbf{R}_{\text{total}} = \mathbf{R} + \mathbf{R}(-1)\mathbf{R} + \mathbf{R}(-1)\mathbf{R}(-1)\mathbf{R} + \cdots = \frac{\mathbf{R}}{1 + \mathbf{R}}. \quad (14)$$

The delay is *inside* $\mathbf{R}$, not beside it. Written with the sub-ocean reflectivity $\mathbf{R}_{\text{sub}}$ made explicit, $\mathbf{R} = E^2 \mathbf{R}_{\text{sub}}$ with $E^2 = e^{2i\omega\eta_w h_w}$ the two-way water phase, and (14) becomes $E^2 \mathbf{R}_{\text{sub}} / (1 + E^2 \mathbf{R}_{\text{sub}})$. A closure written as $\mathbf{R}_{\text{sub}} / (1 + \mathbf{R}_{\text{sub}})$, with the phase dropped, is correct only when the free surface rests directly on the solid — the water thickness then leaves the answer untouched, which is exactly the symptom to test for.

Reconciliation against the Kennett recursion. The global-matrix construction and the Kennett reflectivity recursion are independent routes to the same $\mathbf{R}(p, \omega)$, so their agreement is evidence and their disagreement localises. For the stack above — free surface, ocean, *two* elastic layers and a half-space — the two agree to

Claim	Arbiter	Residual
$\mathbf{R}$ at the top of the water, surface open	Kennett + fluid-solid step	8.8×10^{-16}
$\mathbf{R}$ at the free surface, (14) applied	as above + closure	1.5×10^{-15}
response moves with the water thickness	both sides	1.5

(gate_gmm_marine_stack). Four points of method are worth recording.

Two elastic layers, not one. A single interface cannot exercise interbed multiples, so a recursion that handles one interface correctly and compounds them wrongly would pass a one-layer test.

The PP component only, and that is not a dodge. The two constructions normalise their mode amplitudes differently — the Kennett route by the flux convention $\sqrt{\eta\rho}$, the global matrix by its own eigenvectors. Those differ by a *diagonal* similarity $\mathbf{R} \mapsto \mathbf{C}^{-1}\mathbf{R}\mathbf{C}$, which leaves every diagonal entry exactly invariant. Comparing *PP* is therefore convention-free; comparing the $P \leftrightarrow SV$ conversions would first require that factor, measured elsewhere in this programme to be $i\beta\sqrt{\eta_S} / \alpha\sqrt{\eta_P}$.

The reference depth is a trap. The Kennett recursion references its cumulative reflection at the first *interface*, not at the top of the stack: it phases the layer below each interface, so the leading layer’s own two-way delay is excluded. Handing it the sub-ocean layers directly gives a reflectivity short by one layer’s delay, and because that shifts *P* and *SV* alike it reads as a physics failure rather than a bookkeeping one. A zero-thickness layer of the first solid’s own medium is prepended so that the reflectivity refers to the seabed.

Both sides must attenuate the water identically. Building the water’s vertical slowness from the real velocity while the other side uses the complex attenuative slowness leaves a residual $\omega h_w \eta_w / 2Q$ — proportional to frequency and to water thickness, and nearly independent of the ray parameter. That signature identifies a phase error in the water leg rather than an error in the physics, and was measured at 10^{-5} for $Q = 10^6$ before the two were brought into line.

What this does not yet reach. The reconciliation above is for the plane-wave *reflectivity*. The Green’s-function path of the same construction — the one that supplies the stratified propagator to the scattering solver — does *not* carry the ocean as a finite layer: its response is independent of the water thickness to six significant figures, which by the argument under (14) means the ocean is being treated as a half-space and there is no free surface in it at all. The seabed interface itself is present and correctly parameterised. Closing that gap is a matter of carrying the water phase and the pressure-release closure into the Green’s-function sweep, and it is not done here.

6 Implementation and validation

Object	Where	Note
layer eigenvectors $\mathbf{E}_d, \mathbf{E}_u$	layer_matrix.py	<i>P</i> – <i>SV</i> 4×2 ; <i>SH</i> 2×1
symplectic form $\mathbf{J}$	source_jump.py	J_PSV, J_SH
$\mathbf{E}^{-1}$ in closed form	source_jump.py	symplectic_inverse
unit-jump amplitudes	source_jump.py	unit_jump_amplitudes
source placement (13)	riccati_solver.py	compute_source_vector
Riccati sweep (4)	riccati_solver.py	homogeneous and particular together
layered Green’s function	layered_greens.py	assembles <i>P</i> – <i>SV</i> and <i>SH</i>

The structural claims above are measured, not assumed:

Claim	Residual
$\mathbf{G}$ block-antidiagonal with diagonal blocks, <i>P</i> – <i>SV</i>	4.1×10^{-17}
$\mathbf{G}$ block-antidiagonal with diagonal blocks, <i>SH</i>	0 (exact)
$\mathbf{E}^{-1}$ closed form vs. numerical inverse	2.5×10^{-16}
$\mathbf{E}\mathbf{c} = \mathbf{I}$ for the unit jump	10^{-12}

at ray parameters $p = 0.02, 0.12, 0.25$.

Two traps, recorded because both were hit. The sparsity pattern of $\mathbf{G}$ is $P_{\downarrow} \leftrightarrow P_{\uparrow}$ and $S_{\downarrow} \leftrightarrow S_{\uparrow}$, *not* the literal anti-diagonal; testing the anti-diagonal instead reports a spurious 0.99 failure and invites a hunt for a defect that is not there. And the P – SV form pairs u_x with τ_{xz} and u_z with τ_{zz} with the *same* sign: from Hooke’s law the P column carries $\tau_{xz}/(-i\omega) = -2\mu p\xi$ where the implementation has $+2\mu p\xi$, and likewise on the S column, so a consistent sign on the τ_{xz} row is a basis convention rather than an error — which the symplectic test confirms.

Why the structure is worth testing directly. Equation (9) is a strong constraint that needs no reference implementation: it is a property of the eigenvectors alone. A wrong impedance in any channel destroys it immediately, which makes (9) the cheapest available check on a layer matrix — cheaper, and sharper, than comparing a computed reflectivity against another code.

7 What is distinctive

Two things. The source is treated as a state jump rather than as a body force, which removes the boundary-layer restriction and makes the six-column unit jump the natural object: one solve yields the entire Green’s function. And the decomposition of that jump is obtained from the symplectic structure rather than by inverting a matrix, so what looks like a per-frequency, per-wavenumber 6×6 inversion is in fact a swap and six reciprocals. The global matrix supplies the stratified reference on which the scattering series is built, and both economies apply at every point of that series.